

LAKSHAY LALIA

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EDUCATION

• National Institute of Technology, Hamirpur

B.Tech in Computer Science and Engineering

◦ June 2023 – Present

CGPA: 8.6

EXPERIENCE

• Full Stack Developer Intern

React.js, Python, AWS Lambda

2025

HelioExpect

- Collaborated with UI/UX team to deliver **seamless frontend integrations**, translating **15+ Figma designs** into React.js components; reduced design-to-code turnaround by **30%**.
- Developed and deployed **10+ backend API endpoints** in Python, cutting server response latency by **25%** through serverless architecture optimizations.
- Designed and optimized database schemas and queries, improving data retrieval efficiency by **30%** across core application workflows.

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Java, Python, JavaScript, TypeScript
- **Web Technologies:** React.js, Next.js, Bootstrap, Tailwind CSS
- **Backend and Databases:** Node.js, Express.js, MongoDB, SQL, EJS, REST APIs
- **Tools and Platforms:** Git, GitHub, VS Code, Postman, MySQL, PostgreSQL
- **Core CS:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Cloud Computing, Distributed Systems, Compiler Design, DBMS

PROJECTS

• AstraDB: Query Language Compiler for In-Memory Database

Tools: C++, File I/O, B+ Tree



- Built **lexer and parser** for custom SQL-like query language (**AstraQL**) in C++, supporting **10+ SQL operations**; reduced parse errors by **90%** and achieved query execution under **15ms**.
- Implemented grammar rules, token definitions, AST generation, syntax validation, error handling, and integrated with team's **B+ Tree**-based engine via modular OOP design.

• Chirp: Full-Stack Real-Time Chat Application

Tools: React, React Query v5, Node.js, Express.js, MongoDB, Socket.io



- Developed a robust full-stack chat platform with **Google OAuth 2.0** and **JWT** auth; supports **solo and group chats** (up to **100 members**) with admin controls for member management, group creation, and deletion.
- Integrated **Socket.io** with **10+ custom events** for real-time messaging, **typing indicators**, and online presence; architected **30+ REST API endpoints** across auth, user, message, and group modules using a service-controller-route pattern.
- Designed **React** frontend with **React Query v5** for caching and optimistic updates; added **Cloudinary** image uploads (**10MB** cap via Multer) for avatars and in-chat media sharing.

• Vibble: Backend for Hybrid Video and Microblogging Platform

Tools: Node.js, Express.js, MongoDB, REST API



- Defined **20+ scalable REST API endpoints** supporting video uploads up to **500MB** and microblogging features including posts, likes, comments, and subscriptions.
- Implemented **JWT-based authentication** and role-based authorization across **4 user roles**; load tested with **200 concurrent users** achieving **avg 3.7ms** response time and **p95 under 6ms** with **0% error rate**.

ACHIEVEMENTS

- Solved **750+** problems across LeetCode, Codeforces, and GFG [Click here](#)
- Cleared **Round 2** of **HackWithInfy 2026** by Infosys, selected from **50,000+** applicants nationwide.
- **Runner-Up**, HackSecure'26 Mobile Device Security Hackathon, competing against **150+ teams**.

LEADERSHIP & VOLUNTEERING

• Student Coordinator, CSEC Society

Organizer & Leader

2024 – Present

NIT Hamirpur

- Organized **Hack 5.0** with **1500+ participants**; conducted **5+ CP events** and coding workshops.

• Volunteer, Prayas Society

Teaching Volunteer

2023 – Present

NIT Hamirpur

- Volunteered to teach underprivileged school children free of cost under Literacy Mission NITH.